


# Secondary 4 (Grade 10) – GEP Practice

# 2019 & 2020

## Contest Problems with Full Solutions

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# About the Authors:

**Henry Ong** is the Founder of SASMO and President of Singapore International Mastery Contest Centre (SIMCC). He conducts Math Olympiad classes, not only in Singapore but also around the 20 countries that have joined SASMO. He now travels round the world giving out awards and scholarships to students and teachers. He is now promoting new contests, International Scholastic Trust Teachers' Institute (ISTTI) and International Junior Honor Society (IJHS).

In February 2020, Henry was appointed Co-Chair of the Council Administration of Cambodian Mathematical Society (CA_CMS). Henry will share and use his great wealth of experience, expertise, and international networks to contribute effectively to: (1) promoting STEM: Science, Technology, Engineering, and Mathematics; (2) organizing and undertaking multilateral partnership programs; and (3) strengthening and improving the public-private partnership for development of STEM, research and development, and research and innovation as well as teacher professional development in Cambodia with relevant stakeholders in close collaboration and consultation with relevant leadership and management of Ministry of Education, Youth and Sport, especially Cambodian Mathematical Society (CMS), New Generation Pedagogical Research Center (NGPRC), National Institute of Education(NIE) and Teacher Training Department (TTD).

**Merlan Nagidulin** is the Academic Director for Math and Science Olympiad and General Manager of SIMCC. Merlan graduated with a Bachelor of Science, Honours in Mathematical Sciences from Nanyang Technological University. Merlan won 2 gold medals in Kazakhstan National Math Olympiad and was awarded a bronze medal in the most prestigious International Math Olympiad (IMO Slovenia 2006).

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# International Scholastic Trust President's Message

Dear SIMCC contestants, parents, teachers, and partners,

I hope all of you are safe and healthy. We have all gone through so much within the past year, and we should rejoice - we are all alive and well. Despite all the groom and predictions of the worst recession to hit us very soon, we at SIMCC, are expanding globally – adding another 14 country partners to a total of 35 countries and territories within the last 10 months, and setting up sales offices in Cambodia and India, in addition to taking over the ICAS competitions and staff from the University of New South Wales Global Office based in Hong Kong and Macau.

Two years, Sophie Koh, our Senior Director of IT, Events and Operations left Singapore Polytechnic as a Senior Lecturer of IT to join us. She has almost digitally transformed most of our operations and contest systems. This will provide valuable big data in formative and summative assessments for teachers to measure their students and inform their teaching. Parents and students will also receive Detailed Performance Reports (DPR) in our Member Development Portal (MDP) to improve their Academic Skills. Teachers and their schools will be able to download Students' Analytical Performance Reports (SAPR). SIMCC together with our partners will be conducting more Webinars to prepare teachers and parents to use our contests and data to arm their wards with Skills and Recognition for SUCCESS.

From February 16 to April 30, 2021, we are offering the 1st Singapore Math Global Assessment for grades 1 to 11 students. We want to open this up to every country, especially since Singapore Math textbooks are used in schools from over 70 countries. We want to share our rich Singapore Math heritage that has propelled Singapore students to the top of PISA rankings since the 1990s. We also have excellent teachers, Learning Management System (LMS), Math textbooks, assessment books, as well as teacher training to help improve math education worldwide. Reach out to us for help, if you want your students to become Mathematics Heroes!

The global winners of SINGA Math Assessments are invited for the 1st SINGA Math Global Finals, in their home countries on September 11, 2021. The prize for the Overall Champion of grades 1 to 10/11 is a S$600 competition package to compete in the STEAM AHEAD International Junior Mathematics Olympiad (IJMO) to be held in Bali, Indonesia from December 3 to 7, 2021. The prize for Overall Runner-up is S$250 and 2nd Runner-up is S$150, offering a total prize award of S$10,000. Winners of Gold (10%) and Silver (15%) are invited to compete in STEAM AHEAD IJMO 2021.

From 2021, we will be offering most of our contests to grade 12 students – SASMO, DrCT, SIMOC, and STEAM AHEAD. This will pave the way for Grades 12 SIMCC students to apply for SIMCC and STS's US$270,000 endowed 2 scholarships for 4-year undergraduate studies at Southern Illinois University (SIU), Carbondale, Illinois, USA in STEM related fields. SIU has also given SIMCC over 30 SIU International Student Tuition Grants to award to top SIMCC students to pursue their university education. SIMCC will setup our own network for SIMCC scholars at SIU. This will help us expand SIMCC contests in North America and generate revenue for us to fund more scholarships.

Best Regards,
Henry Ong, President

# Competition and Prizes

SASMO is devoted and dedicated to bringing a love for Mathematics to students. Unlike most Math Olympiad Competitions, SASMO caters not only to students in the top 5% but to the top 40% instead. It aims to arouse students' interest and enthusiasm for mathematical problem solving, develop mathematical intuition, reasoning and logical thinking, as well as creative and critical thinking. In addition, this can help improve the students' math grades because they can apply problem-solving strategies learnt during the training to their daily school mathematics.

History:

Created in 2006, SASMO is one of the largest Math Olympiads in the Asian region. We have expanded the competition to provide an International platform for students from Primary 2 to Secondary 4, with differentiated contest papers for every level. SASMO awards medals and certificates to the top 40% of participants.

Calculators are not permitted

When a problem introduces a more advanced concept, all necessary definitions are included.

Awards:

Each participant receives a Certificate of Participation or an award certificate for winners below. Each of the top 8%, 12% and 20% of all participants receives a Gold, Silver or Bronze medal and certificate respectively. Each student who achieves a Perfect Score of 85 points receives a Perfect Score certificate, Gold medal and $100.

# Introduction

For Students Taking the Math Olympiad Challenge

Congratulations. You have embarked on a journey of scholarship. Competitions like SASMO open many doors for you. Firstly, you learn new and interesting approaches to problem-solving and new topics. Next, you will meet talented students from other schools as you attend trainings and competitions. You build your endurance to "puzzle" out challenging problems and build your reputation as a problem solver. Finally, you will be exposed to various international competitions and scholarship opportunities. Here in Singapore, you increase your chances of getting into a top school via Direct School Admissions (DSA) and entry into the Gifted Education Programme as you compete regularly in high level competitions.

This book is written for the participants in the Singapore and Asian Schools Math Olympiads (SASMO). It helps students to prepare well for the contest and develop higher-order thinking. All problems are designed to help students develop the ability to think mathematically, rather than to teach more advanced or unusual topics. The fun is in how you can see patterns and ways of solving each problem in non-technical ways even though you have not learnt the topic yet!

In addition to the contest problems, the reader is provided with a list of familiar mathematical terms, as well as a review of some of the topics that are likely to be tested in the Olympiad. The book also contains some solved examples to provide different problem-solving techniques, and to familiarize the participant with different types of Olympiad questions. It is advised that the reader spends appropriate time studying these questions and solutions, as they will assist in tackling actual Olympiad problems.

How to Use This Book: Practice daily for 15 minutes per hour instead of 4 hours of learning once a month. Your mind needs to absorb each new thought, and constant practice allows frequent review of previously learned concepts and skills. Together, you can remember many new problem-solving approaches. Try to spend 10 or 15 minutes daily doing two or three problems. This approach should help you minimize the time needed to develop the ability to think mathematically.

Whether you solve a problem quickly or if you are confused, it is worth studying the solutions in this book, because often they offer unexpected insights that can help you understand the problem more fully. After you have invested time – trying to solve each problem any way you can, reviewing our solutions is very effective. Many of the problems in this book can be solved in more than one way. There is always a single answer, but there can be many paths to that answer. Once you solve a problem, go back and see if you can solve it by another method. Then check our solutions to see if any of them differ from yours.

Enjoy working on these challenges and you will soon be in a different league from your peers who have not taken any international competition. We look forward to inviting you if you are a bronze, silver, gold or perfect score medallist for further training.

# Problem Solving Procedure

You may go through several phases when solving a problem such as trying to understand the problem, working on a specific approach (planning and attempting), getting stuck and trying to get unstuck, critically examining solutions or communicating. The work may involve going back and forth between these different phases of problem solving.

In solving any problem, it helps to have a working procedure. You might want to consider this four-step procedure: Understand, Plan, Try It, and Look Back.

**Understand**

Before you can solve a problem, you must first understand it. Read and re-read the problem carefully to find all the clues and determine what the question is asking you to find. What is the unknown? What is the data? What is the condition?

**Plan**

Once you understand the question and the clues, it's time to use your previous experience with similar problems to look for strategies and tools to answer the question. Do you know a related problem? Look at the unknown! And try to think of a familiar problem having the same or a similar unknown?

**Try It**

After deciding on a plan, you should try it and see what answer you come up with. Can you see clearly that the steps are correct?
But can you also prove that the steps are correct?

**Are you feeling stuck?**

Many different approaches can be tried to get unstuck. One approach is to try working a simpler version of the problem, and use the solution to the problem to get insights that are useful in solving the original problem. In the next chapter, we will show some common solving approaches.

If you are discouraged after a few failed attempts, read this quote from the famous scientist, Thomas Edison. An assistant asked, "Why are you wasting your time and money? We have had failure after failure, almost a thousand of them. Why do you continue to pursue this impossible task?" Edison said, "We haven't had a thousand failures, we've just discovered a thousand ways to not invent the electric bulb."

**Look Back**

Once you've tried a problem and found an answer, go back to the problem and see if you've really answered the question. Sometimes it's easy to overlook something. If you missed something check your plan and try the problem again.
Can you check the result?
Can you check the argument?
Can you derive the result differently?
Can you see it at a glance?

# Problem Solving Strategies

## 1. Change the representation

Using a wrong representation may make a problem impossible to solve.
Strategies of changing representation include drawing a picture and acting it out.

**DRAW A PICTURE:** By drawing a picture, and visualizing the problem information using it, you will have a clearer understanding of the problem and it will help you to come up an approach to solve the problem that you might not be able to see otherwise.

**ACT IT OUT:** We are better at thinking in terms of concrete objects and situations than in terms of abstract concepts. If we can act out the situation described in a word problem, we are able to understand the problem better and we may be able to come up with a solution. To do this, we need to use real materials that are easily available to us. Examples can be pencils, coins and other objects we have in the classroom.

## 2. Make an Organized List or a Table

**ORGANIZED LIST**: Making an organized list allows you to clearly examine data. It can help you in ensuring that you are looking at all the relevant information. It will also allow you to see patterns in the data easily and to come to correct conclusions.

**MAKE A TABLE:** Making a table allows you to clearly examine data. It can help you in ensuring that you are looking at all the relevant information. It will also allow you to see patterns in the data easily and to come to correct conclusions.

### 3. Create a Simpler Problem

Sometimes we are not able to solve the problem as it is stated, but we are able to solve a similar problem that is similar in some way. For example, the simpler problem may use simpler numbers. Once we solve one or more simpler problems, we may understand the approach that can be used to solve the problems of similar type and may be able to solve the problem that has been given to us.

### 4. Use Logical Reasoning

Logical reasoning is useful in mathematics problems in various ways. It can be used to eliminate incorrect choices. It can also sometimes be used to conclude the answer directly.

### 5. Guess and Check

"Guess and Check" strategy can be used on many problems. If the number of possible answers is small, one can use this strategy to come up with the answer very quickly. In some other cases where the number of possible answers is not small, one may still be able to make intelligent guesses and come up with the answer.

### 6. Working Backwards

Sometimes, it is easier to start with information at the end of the problem and work backwards to the beginning of the problem than the other way around.

# SASMO 2019 Secondary 4 (Grade 10) Contest Questions

**Section A (Correct answer – 2 points| No answer – 0 points| Incorrect answer – minus 1 point)**

## Question 1

One-fourth of $16^{40}$ is equal to

A. $4^{40}$
B. $16^{10}$
C. $4^{79}$
D. $4^{41}$
E. None of the above

## Question 2

The following calculations lead to the conclusion that $\pi > 2019$.

Let $x = \pi = 3.14159\ldots$ (1)

$\frac{2x+38}{x-1000} < 4$ (2)

$2x + 38 < 4(x - 1000) = 4x - 4000$ (3)

$2x > 4038 \quad \Rightarrow \quad x > 2019$ (4)

Therefore, $\pi > 2019$ (5)

Which of the following transition steps is the first that contains an error?

A. (1) to (2)
B. (2) to (3)
C. (3) to (4)
D. (4) to (5)
E. None of the above

**Question 3**

The moment of inertia is a measure of an object's tendency to resist rotational acceleration. The moment of inertia $(I)$ of a hollow sphere rotating about an axis through its centre is

$$I = \frac{2}{3}mr^2,$$

where $m$ is its mass and $r$ is its radius. If the moment of inertia of a hollow sphere becomes half of the original and its mass is doubled, what will happen to its radius?

A. Halved
B. Doubled
C. Reduced to $\frac{1}{4}$ of the original
D. Reduced to $\frac{1}{8}$ of the original
E. None of the above

**Question 4**

Which of the following statements is not true?

A. $\tan\theta > 0$ for $0° < \theta < 90°$
B. $\sin\theta < 0$ for $270° < \theta < 360°$
C. $\cos\theta > \sin\theta$ for $0° < \theta < 90°$
D. $\cos\theta < 0$ for $90° < \theta < 180°$
E. None of the above

**Question 5**

How many prime numbers $x$ satisfy the following inequality?

$$\frac{x+15}{2} \geq \frac{3x-7}{4} + 1 > 5\frac{1}{2}$$

A. 25
B. 7
C. 8
D. 12
E. None of the above

**Question 6**

Let $m$ be a positive integer such that the sequence below is an arithmetic progression. How many terms are there in the list below?

$$\{6, 13, 20, 27, \ldots, m+7\}$$

A. $\frac{m+8}{7}$
B. $\frac{m+1}{7}$
C. $m+1$
D. $\frac{m+7}{7}$
E. None of the above

**Question 7**

A positive integer $n$ gives a remainder of 3 when divided by 5. Which of the following numbers is divisible by 5?

A. $n+3$
B. $n-2$
C. $5n+3$
D. $5n+2$
E. $2n+4$

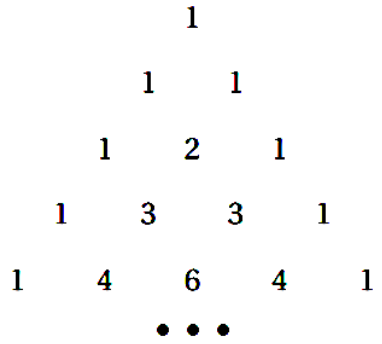
**Question 8**

The value of $\sqrt{18}+\sqrt{27}$ is between two consecutive positive integers. Which of following are the values of these 2 integers?

A. 9 and 10
B. 8 and 9
C. 10 and 11
D. 7 and 8
E. None of the above

**Question 9**

The coefficients in the expansion of $(x+y)^n$ are related to the numbers in the "Pascal triangle" on the right.




For instance, $(x+y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$. Its coefficents $1, 3, 3$ and $1$ are the numbers in the 4th row of the Pascal triangle.

What is the sum of the coefficients of $(x+y)^8$?

A. 128
B. 256
C. 512
D. 127
E. None of the above

**Question 10**

In a specific week this year, the probability of raining on any given day in Singapore is 50%. What is the probability of raining in Singapore on exactly 2 days in that week?

A. $\frac{1}{4}$
B. $\frac{3}{64}$
C. $\frac{21}{64}$
D. $\frac{21}{128}$
E. None of the above

**Question 11**

Find the remainder when $19^5 - 19$ is divided by $30$.

A. 0
B. 1
C. 9
D. 18
E. None of the above

**Question 12**

In which figure will the number of black circles in the figure be 111 more than the number of white circles in the same figure?

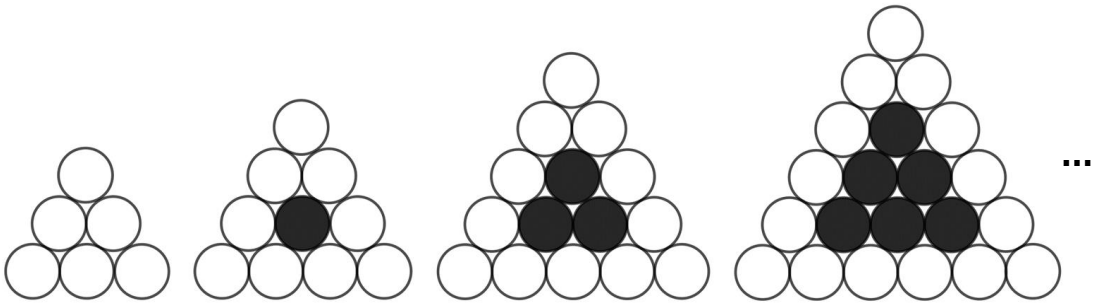


Figure 1 Figure 2 Figure 3 Figure 4

A. Figure 18
B. Figure 19
C. Figure 20
D. Figure 12
E. None of the above

**Question 13**

What is the last digit of the product $549^{123} \times 549^{123}$?

A. 1
B. 9
C. 3
D. 6
E. None of the above

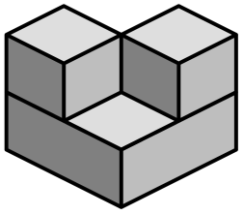
**Question 14**

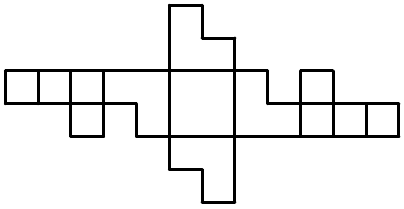
What is the 2019th digit after the decimal point of the decimal form of $\frac{2019}{7}$?

A. 5
B. 8
C. 7
D. 1
E. None of the above

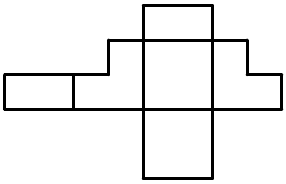
**Question 15**

Which of the nets below can be folded to form the figure on the right?

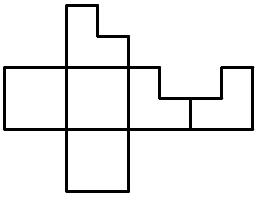




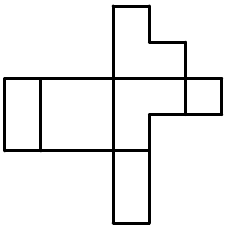
A



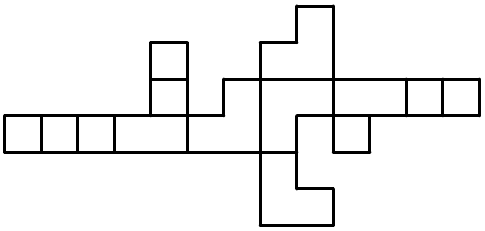
B



C



D



E

**Section B (Correct answer – 4 points| Incorrect or No answer – 0 points)**

When an answer is a 1-digit number, shade "0" for the tens, hundreds and thousands place.
*Example: if the answer is 7, then shade 0007*
When an answer is a 2-digit number, shade "0" for the hundreds and thousands place.
*Example: if the answer is 23, then shade 0023*
When an answer is a 3-digit number, shade "0" for the thousands place.
*Example: if the answer is 785, then shade 0785*
When an answer is a 4-digit number, shade as it is.
*Example: if the answer is 4196, then shade 4196*

## Question 16

A number $A = \underbrace{20192019\ldots2019}_{n\ times\ 2019}$ is formed by writing 2019 next to each other $n$ times. What is the least possible positive value of $n$ such that $A$ is divisible by 9?

## Question 17

A list contains 5 positive integers that satisfy the following conditions:

- The average of the five numbers is 86.
- The median is 99.
- The largest number is 12 more than 20 times the smallest number.

What is the maximum possible value of the largest number?

**Question 18**

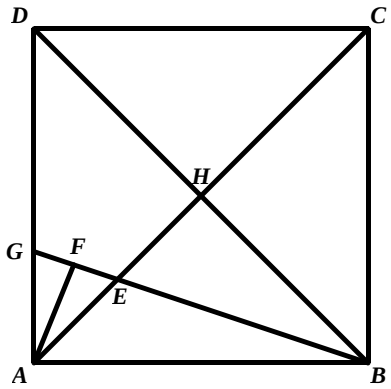
How many 4-digit perfect squares start with two 9s?

**Question 19**

How many digits are in the number $3^3 \times 8^{14} \times 25^{18}$?

**Question 20**

In the diagram, $ABCD$ is a square, and $H$ is the intersection point of its diagonals. Points $B$, $E$, $F$ and $G$ are collinear such that point $G$ is on side $AD$, point $E$ on diagonal $AC$, $AF$ is the bisector of $\angle CAD$ and $\angle AFB = 90°$. If $EH = 13\sqrt{2}$, find the length of $AG$.



**Question 21**

Find the sum of all possible real values of $n$ that satisfy

$|n^2 + 21 - 10n| = |63 - 2n - n^2|$.

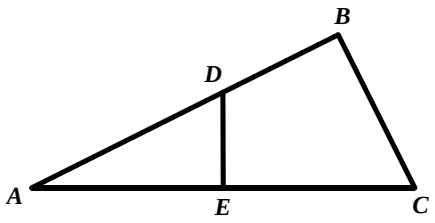
**Question 22**

It is given that $371n + 25$ is a four-digit perfect square, where $n$ is a positive integer. Find the value of $n$.

**Question 23**

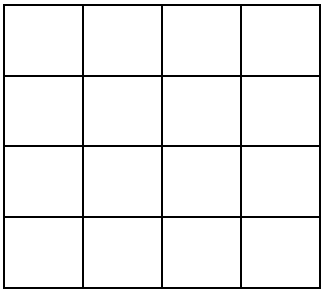
In the diagram, $ABC$ is a right-angled triangle with $AB = 15$ and $BC = 8$. Point $D$ is on side $AB$ such that $DB = 6$. Point $E$ is on hypotenuse $AC$ such that $DE$ is perpendicular to $AC$. Find the length of $BE$.

(Round off your answer to the nearest integer.)



**Question 24**

How many ways to fill a $4 \times 4$ grid with 0 or 1 only such that the sum of every row and column is even?



**Question 25**

In the following cryptarithm, all the different letters stand for different digits.

$$\begin{array}{cccccc} & & C & O & O & L \\ + & & M & A & T & H \\ \hline & S & A & S & M & O \\ \hline \end{array}$$

Given that M = 4, find the value of S + A + S + M + O.

**END OF PAPER**

# SASMO 2020 Secondary 4 (Grade 10) Contest Questions

**Section A (Correct answer – 2 points| No answer – 0 points| Incorrect answer – minus 1 point)**

## Question 1

Find the value of the following.

$$2020 \times 999 - 9999 \times 202 + 9999$$

A. 8181
B. 2020
C. –1818
D. 4047777
E. None of the above

## Question 2

How many digits are there in $5^{2020} \times 4^{1008}$?

A. 2020
B. 1008
C. 2016
D. 2019
E. None of the above

## Question 3

A standard 6-sided dice is rolled twice. Which of the following events has the highest probability of occurrence?

A. The second number is twice the first.
B. The second number is not greater than the first.
C. At least one number is greater than 3.
D. The sum of two numbers is a prime number.
E. None of the above

## Question 4

Each of six countries brought a team of 5 players to a tennis tournament. Each pair of players from different countries played once. How many games were there in the tournament?

A. 250
B. 360
C. 375
D. 480
E. None of the above

## Question 5

A "Nicely Seven" is a two-digit positive integer such that it has only two factors and the sum of the factors is a multiple of 7. How many Nicely Seven numbers are there?

A. 8
B. 13
C. 3
D. 4
E. None of the above

## Question 6

Given that $a = 4^{\frac{1}{3}} - 2^{\frac{1}{3}} + 1$, what is the value of $\left(\frac{3-a}{a}\right)^6$?

A. $4$
B. $2$
C. $64$
D. $\left(4^{\frac{1}{3}} - 2^{\frac{1}{3}}\right)^2$
E. None of the above

## Question 7

Find the greatest prime number $x$ which satisfies the following inequality.

$$\left|\frac{1}{57-x}\right| < \frac{5}{x^2}$$

A. 13
B. 11
C. 59
D. 61
E. None of the above

## Question 8

The number 4 can be expressed as the sum of the digits 1 or 2 in five different ways as shown below.

$4 = 1 + 1 + 1 + 1$

$4 = 1 + 1 + 2$

$4 = 1 + 2 + 1$

$4 = 2 + 1 + 1$

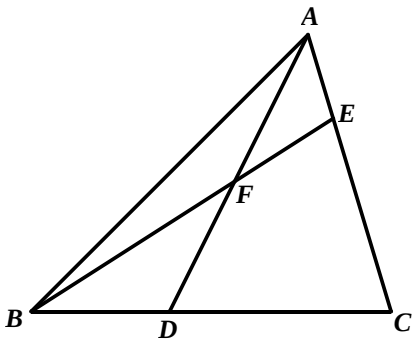
$4 = 2 + 2$

How many ways can the number 9 be expressed as the sum of the digits 1 or 2?

A. 60
B. 55
C. 45
D. 40
E. None of the above

## Question 9

In the diagram, points $E$ and $D$ are on the sides $AC$ and $BC$ such that $AE = \frac{1}{3}AC$ and $BD = \frac{1}{3}BC$. What is the ratio of the area of triangle $AEF$ to the area of triangle $ABC$?




A. $\frac{2}{7}$
B. $\frac{2}{9}$
C. $\frac{2}{13}$
D. $\frac{2}{15}$
E. None of the above

## Question 10

Dennis wrote down all the 3-digit numbers in increasing order on a piece of paper. He used a red pen to write down the even numbers and a blue pen to write down the odd numbers. How many red digits '6' are there on the piece of paper?

A. 185
B. 195
C. 280
D. 900
E. None of the above

## Question 11

It is given that $a, b,$ and $c$ are real numbers that satisfy the following equations.

$$\begin{cases} \dfrac{ab}{a+b} = 4 \\ \dfrac{bc}{b+c} = 6 \\ \dfrac{ac}{a+c} = 12 \end{cases}$$

What is the value of $\frac{abc}{ab+bc+ac}$?

A. 20
B. 2
C. 4
D. $\frac{1}{4}$
E. None of the above

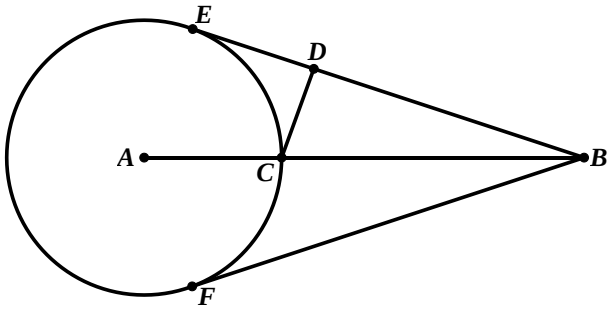
## Question 12

How many pairs of two-digit positive integers $a$ and $b$ are there such that $a^2 + b^2 = 2020$?

A. 0
B. 1
C. 2
D. 4
E. None of the above

## Question 13

In the figure below, the circle is centred at $A$. The segments $BE$ and $BF$ are tangents to the circle. Point $C$ is on $AB$ and on the circumference of the circle as shown. Point $D$ is a point on $BE$ such that $CD$ is perpendicular to $BE$. If $BE = 12$ and $BC = 8$, find the length of $CD$.




A. $\frac{25}{13}$
B. $\frac{24}{13}$
C. $\frac{40}{13}$
D. $\frac{12}{5}$
E. None of the above

## Question 14

Diana, Alexa, Jason, Tom and Michael are Secondary 2 or 4 students. They study in either Silver Oak Secondary School or Grand Mountain Secondary School. It is also given that:

- Tom and Michael are from different schools.
- Diana and Jason go to the same school.
- Three students go to Silver Oak Secondary School and the other two are from Grand Mountain Secondary School.
- Alexa and Michael are from the same grade.
- Jason and Tom study at different levels.
- Three students study in Secondary 2 and the other two students in Secondary 4.

If one of them is Secondary 4 student from Grand Mountain Secondary School, who is that person?

A. Diana
B. Alexa
C. Jason
D. Tom
E. Michael

## Question 15

Thirty identical marbles are to be distributed to 6 boys. Each boy gets at least 3 marbles. How many ways are there to distribute the marbles to the boys?

A. $6188$
B. $15625$
C. $118755$
D. $\frac{30!}{6}$
E. None of the above

## Section B (Correct answer – 4 points| Incorrect or No answer – 0 points)

When an answer is a 1-digit number, shade "0" for the tens, hundreds and thousands place.
*Example: if the answer is 7, then shade 0007*
When an answer is a 2-digit number, shade "0" for the hundreds and thousands place.
*Example: if the answer is 23, then shade 0023*
When an answer is a 3-digit number, shade "0" for the thousands place.
*Example: if the answer is 785, then shade 0785*
When an answer is a 4-digit number, shade as it is.
*Example: if the answer is 4196, then shade 4196*

### Question 16

Find the largest 4-digit multiple of 25 whose sum and product of its digits are both multiples of 25.

### Question 17

What is the value of the following expression?

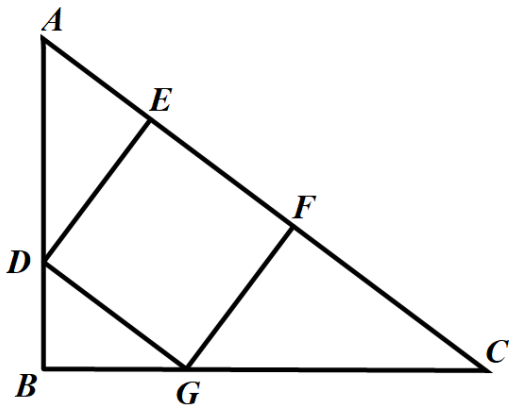
$$\frac{3 \times 111.111}{6 \times 1.001} + \frac{3 \times 222.222}{6 \times 2.002} + \frac{3 \times 333.333}{6 \times 3.003} + \cdots + \frac{3 \times 666.666}{6 \times 6.006}$$

## Question 18

The six-digit number 21A3B8 is divisible by 33. Find the value of A + B.

## Question 19

In the diagram, $\angle ABC = 90°$ and $DEFG$ is a square. If $AE = 8$ and $EF = 12$, find the area of quadrilateral $ACGD$.



## Question 20

What is the greatest 2-digit number that can be written as the sum of 2 different prime numbers in exactly 2 different ways?

## Question 21

What is the smallest positive integer that has exactly six odd divisors and 12 even divisors?

## Question 22

If $(x^2+1)(y^2+1)+9=6(x+y)$, find the value of $x^2+y^2$.

## Question 23

What is the least number of weights required to weigh any objects of integer number of grams from 1 to 35 grams? The weights must be in integer number of grams.

## Question 24

In the following cryptarithm, all the different letters stand for different digits.

$$\begin{array}{cccccccc} & C & I & R & C & L & E \\ & C & I & R & C & L & E \\ + & C & I & R & C & L & E \\ \hline & S & P & H & E & R & E \end{array}$$

Find the value of the sum C + I + R + C + L + E.

## Question 25

Peter has seven different books of different subjects to be placed on a single-decked shelf. He does not want to place the Physics book next to the Biology one and the Geometry book next to the Chemistry one. In how many ways can he place all his books?

**END OF PAPER**

# Solutions to SASMO 2019 Secondary 4 (Grade 10)

## Question 1

$$\frac{1}{4} \times 16^{40} = \frac{(4^2)^{40}}{4} = \frac{4^{80}}{4} = \mathbf{4^{79}}$$

Answer: **(C)**

## Question 2

If $\frac{2x+38}{x-1000} < 4$, then $2x + 38 > 4(x - 1000)$ as $(x - 1000) < 0$.

Therefore, the transition step from **(2) to (3)** is wrong.

Answer: **(B)**

## Question 3

Since the moment of inertia of a hollow sphere becomes half of the original and its mass is doubled, then

$$I_{new} = \frac{I}{2} \text{ and } m_{new} = 2m$$

$$r_{new} = \sqrt{\frac{3I_{new}}{2m_{new}}} = \sqrt{\frac{\frac{3I}{2}}{4m}} = \sqrt{\frac{1}{4} \times \frac{3I}{2m}} = \frac{1}{2} \times \sqrt{\frac{3I}{2m}} = \frac{1}{2}r$$

The new radius becomes **half of the original**.

Answer: **(A)**

## Question 4

**$\cos\theta$ is not always bigger than $\sin\theta$ for $0° < \theta < 90°$**. For example, $\sin 60° = \frac{\sqrt{3}}{2}$ and $\cos 60° = \frac{1}{2}$.

Answer: **(C)**

**Question 5**

$$\frac{x+15}{2} \geq \frac{3x-7}{4} + 1 \Leftrightarrow 2x + 30 > 3x - 7 + 4 \Leftrightarrow 33 > x$$

$$\frac{3x-7}{4} + 1 > 5\frac{1}{2} \Leftrightarrow 3x - 7 + 4 > 22 \Leftrightarrow 3x > 25 \Leftrightarrow x > 8$$

**Seven** primes satisfy the inequality $8\frac{1}{3} < x \leq 33$: 11, 13, 17, 19, 23, 29 and 31.

Answer: **(B)**

**Question 6**

The common difference of the arithmetic progression is 7. Therefore, there are

$$\frac{(m+7)-6}{7} + 1 = \boldsymbol{\frac{m+8}{7}} \text{ terms.}$$

Answer: **(A)**

**Question 7**

If $n$ gives a remainder of 3 when divided by 5, then $n - 3$ is divisible by 5.

$$2n + 4 = 2n - 6 + 6 + 4 = 2(n-3) + 10$$

Since $2(n-3)$ and 10 are divisible by 5, then $\boldsymbol{2n+4}$ is also divisible by 5.

Answer: **(E)**

**Question 8**

$$\left(\sqrt{18} + \sqrt{27}\right)^2 = 18 + 27 + 2 \times \sqrt{18} \times \sqrt{27} = 45 + 2 \times \sqrt{9 \times 9 \times 2 \times 3} = 45 + 18 \times \sqrt{6}$$

Since $2 < \sqrt{6} < 3$,

$$45 + 18 \times 2 < 45 + 18 \times \sqrt{6} < 45 + 18 \times 3$$

$$81 < 45 + 18 \times \sqrt{6} < 99 < 100$$

$$81 < \left(\sqrt{18} + \sqrt{27}\right)^2 < 100$$

$$9^2 < \left(\sqrt{18} + \sqrt{27}\right)^2 < 10^2$$

$$\boldsymbol{9 < \sqrt{18} + \sqrt{27} < 10}$$

Answer: **(A)**

## Question 9

To get the sum of coefficients, we substitute $x = y = 1$, i.e., $(1+1)^3 = 1+3+3+1 = 8$.

The sum of the coefficients of $(x+y)^8$ is

$$(1+1)^8 = 2^8 = \mathbf{256}.$$

Answer: **(B)**

## Question 10

The probability of rain each day is $\frac{1}{2}$ and the probability of no rain is also $\frac{1}{2}$. There are $\frac{7\times6}{2} = 21$ ways to choose on which 2 days out of the 7 possible days that there will be a rain. So, the probability is

$$21 \times \left(\frac{1}{2}\right)^2 \times \left(\frac{1}{2}\right)^5 = \frac{\mathbf{21}}{\mathbf{128}}.$$

Answer: **(D)**

## Question 11

$19^5 - 19 = 19 \times (19^4 - 1) = 19 \times (19^2 - 1) \times (19^2 + 1) = 19 \times (19-1) \times (19+1) \times (19^2+1) = 19 \times 18 \times 20 \times (19^2+1) = 360 \times 19 \times (19^2+1)$ which is divisible by 30. Hence the remainder is **0**.

Answer: **(A)**

## Question 12

From the table below, we can derive formula for the number of white and black circle in each figure.

| **Figure** | **1** | **2** | **3** | **4** | **n** |
|---|---|---|---|---|---|
| Number of white circles | $6 = 2 \times 3$ | $9 = 3 \times 3$ | $12 = 4 \times 3$ | $15 = 4 \times 3$ | $(n+1) \times 3$ |
| Number of black circles | $0$ | $1 = \frac{1 \times 2}{2}$ | $3 = \frac{2 \times 3}{2}$ | $6 = \frac{3 \times 4}{2}$ | $\frac{(n-1) \times n}{2}$ |

$$\frac{(n-1) \times n}{2} - (n+1) \times 3 = 111$$

$$n^2 - n - 6n - 6 = 222$$

$$n^2 - 7n - 228 = 0$$

$$(n-19)(n+12) = 0$$

$$n = 19.$$

In **Figure 19**, the number of black circles in the figure will be 111 more than the number of white circles.

Answer: **(B)**

## Question 13

The last digit of $549^{123} \times 549^{123} = 549^{246}$ is the same as the last digit of $9^{246}$.

Based on the table below, the last digits of the powers of 9 have pattern length of 2.

| Powers of 3 | Last digit |
|---|---|
| $9^1$ | 9 |
| $9^2$ | 1 |
| $9^3$ | 9 |
| $9^4$ | 1 |

Since 246 is an even number, then $9^{246}$ ends in 1 and $549^{246}$ also ends in **1**.

Answer: **(A)**

## Question 14

The fraction $\frac{2019}{7}$ is equal to $288.\overline{428571}$, where $\overline{428571}$ means that the series of digits '$428571$' is repeated indefinitely. The length of the repeated series is 6. Since $\frac{2019}{6}$ gives a remainder of 3, then this 2019th digit is the 3rd digit in the repeated series, which is **8**.

Answer: **(B)**

## Question 15

If the area of a small square on the surface of the figure is 1, then the surface area of the figure is at least 22. Hence options B, C and D are not possible. Option A is also not possible. The answer is **option E**.

Answer: **(E)**

## Question 16

If a number is divisible by 9, then the sum of its digits is also divisible by 9.

The sum of the digits of 2019 is $2 + 0 + 1 + 9 = 12$. The closest multiple of 9 is 36. Hence the least possible value of $n$ is **3** and $A = 201920192019$.

Answer: **3**

## Question 17

Let the 5 positive integers be $a, b, c, d$ and $e$, where $a \geq b \geq c \geq d \geq e$.

From the given, $c = 99$ is the median.

Also, $a = 20e + 12$ and $\frac{a+b+c+d+e}{5} = 86$. We have

$$86 \times 5 = 430 = a + b + c + d + e = (20e + 12) + b + 99 + d + e$$

$$319 = 21e + b + d \quad (1)$$

For $a$ to be the largest, $e$ must be as large as possible since $a = 20e + 12$.

For $e$ to be the largest, $b$ and $d$ must be as small as possible.

The smallest possible value of $b$ is 99 since $b \geq c = 99$.

The smallest possible value of $d$ is $e$ since $d \geq e$

Substitute $b = 99$ and $d = e$ into the equation (1)

$$319 = 21e + 99 + e = 22e + 99$$

$$319 - 99 = 220 = 22e$$

Hence $e = 10$ and $a = 20e + 12 = 200 + 12 = \mathbf{212}$.

Answer: **212**

## Question 18

Since $100^2 = 10000$, the largest 4-digt perfect square is $99^2 = 9801$. Thus, there are **no** 4-digit perfect squares that start with two 9s.

Answer: **0**

## Question 19

$$3^3 \times 8^{14} \times 25^{18} = 3^3 \times (2^3)^{14} \times (5^2)^{18} = 3^3 \times 2^{42} \times 5^{36} = 3^3 \times 2^6 \times 2^{36} \times 5^{36}$$
$$= 27 \times 64 \times (2 \times 5)^{36} = 1728 \times 10^{36} = 1728\underbrace{00\ldots0}_{36\ 0s}$$

There are $4 + 36 = \mathbf{40}$ digits in $3^3 \times 8^{14} \times 25^{18}$.

Answer: **40**

## Question 20

Triangle $AGE$ is an isosceles triangle with $AG = AE$ since $AF$ is the bisector and the height. Also, triangle ABH is an isosceles right triangle.

Let the length of $AE$ be $a$. Then $HA = HB = 13\sqrt{2} + a$ and $AB = \left(13\sqrt{2} + a\right)\sqrt{2}$.

$$\angle EBH = 90° - \angle HEB = 90° - \angle AEF = \angle FAE = \frac{45°}{2}$$

$$\angle ABE = 45° - \angle EBH = 45° - \frac{45°}{2} = \frac{45°}{2}$$

Therefore, BE is the bisector of $\angle ABH$. Using Bisector Theorem in the triangle $ABH$

$$\frac{AE}{HE} = \frac{AB}{BH} \ \ or \ \ \frac{a}{13\sqrt{2}} = \frac{\left(13\sqrt{2} + a\right)\sqrt{2}}{13\sqrt{2} + a} = \sqrt{2} \ \ \Rightarrow a = 13\sqrt{2} \times \sqrt{2} = \mathbf{26} = AE = AG.$$

Answer: **26**

## Question 21

Factorise both sides of the equation as

$$|(n-7)(n-3)| = |(n+9)(n-7)|$$

Obviously, $n = 7$ is a solution.

Divide both side by $(n-7)$, we get

$$|(n-3)| = |(n+9)|$$

$|(n-3)| = |(n+9)|$ implies $n-3 = n+9$, which does not have any solution, and $n-3 = -(n+9)$ which has one solution $n = -3$.

The sum of all possible real values of $n$ is $7 + (-3) = \mathbf{4}$.

Answer: **4**

## Question 22

Let $371n + 25 = a^2$ where $31 < a < 100$

$$371n = a^2 - 25 = a^2 - 5^2 = (a-5)(a+5)$$

$$(a-5)(a+5) = 371n = 53 \times 7n$$

Hence either $(a-5)$ or $(a+5)$ is divisible by 53 which is a prime number. Since $a < 100$, there are only 3 cases:

$(a-5) = 53$ or $a = 58$

$(a+5) = 53$ or $a = 48$

$(a+5) = 106$ or $a = 101$

Only $a = 58$ has integer solutions. Thus,

$$n = \frac{a^2 - 25}{371} = \frac{58^2 - 25}{371} = \mathbf{9}$$

Answer: **9**

## Question 23

By Pythagoras Theorem

$$AC = \sqrt{AB^2 + BC^2} = \sqrt{15^2 + 8^2} = 17$$

Triangle $ABC$ and $AED$ are similar since $\angle BAC = EAD$ and $\angle DEA = \angle CBA$. Therefore,

$\frac{AD}{DE} = \frac{AC}{BC}$ or $\frac{9}{DE} = \frac{17}{8} \Rightarrow DE = \frac{72}{17}$

$\frac{AD}{AE} = \frac{AC}{AB}$ or $\frac{9}{AE} = \frac{17}{15} \Rightarrow AE = \frac{135}{17}$

$EC = AC - AE = 17 - \frac{135}{17} = \frac{154}{17}$

By Pythagoras Theorem,

$$DC = \sqrt{DB^2 + BC^2} = \sqrt{6^2 + 8^2} = 10\,.$$

Quadrilateral $DECB$ is cyclic since $\angle DEC + \angle DBC = 90° + 90° = 180°$.

By Ptolemy's theorem in $DECB$

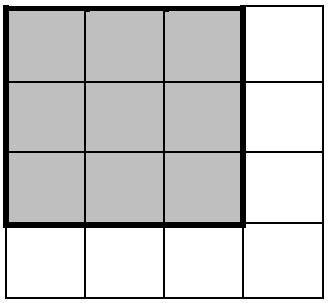
$$EB \times DC = DB \times EC + DE \times BC = 6 \times \frac{154}{17} + \frac{72}{17} \times 8 = \frac{1500}{17}$$

$$EB = \frac{1500}{17} \div DC = \frac{1500}{17} \div 10 = \frac{150}{17} = 8\frac{14}{17}$$

Rounding off $8\frac{14}{17}$ to the nearest integer, we get **9**.

Answer: **9**

## Question 24



Let us refer to the cell in the $a^{\text{th}}$ row and $b^{\text{th}}$ column as the $(a, b)$ cell.

You can fill the shaded $3 \times 3$ grid however you'd like.

Once the shaded cells are filled, then unshaded cells are determined depending on the sum of the numbers in each row and each column of the shaded cells.

For example, the sum of the column is 1+1+1=3, then the number below it can only be a 1 for its sum to be even. Use the figure on the right for an illustration of one possible case.

| | | | |
|---|---|---|---|
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 |
| 1 | 1 | 0 | 0 |

So, the number of ways to fill the $4 \times 4$ grid is the same as the number of ways to fill the shaded $3 \times 3$ grid.

Since there are 2 options (1 or 0) to put in each shaded cell, then the total number of ways to fill the shaded $3 \times 3$ grid is $2^9 = \mathbf{512}$.

Answer: **512**

## Question 25

It is clear that S = 1. To find the value of S + A + S + M + O, we need to find the values of A and O.

If $O = 0$, then $T = 9$ and $M = 0$ or $T = M$ which is impossible. Therefore, $O \neq 0$.

In the tens place addition,

$$O + T = 14, 13, 4 \text{ or } 3$$

Case 1: $O + T = 3$ has 1 possible solution as $O = 3$ and $T = 0$.

| $T$ | $O$ | $A$ | $C$ | **Remarks** |
|---|---|---|---|---|
| 0 | 3 | 8 | None | No possible value for $C$ |

Case 1 is impossible.

Case 2: If $O + T = 4$, has 1 possible solution as $O = 4$ and $T = 0$.

| $T$ | $O$ | $A$ | $C$ | **Remarks** |
|---|---|---|---|---|
| 0 | 4 | 7 | None | No possible value for $C$ |

Case 2 is impossible.

Case 3: $O + T = 13$, then $O + A = 10$. $O + T = 13$ has 4 possible solutions:

| $T$ | $O$ | $A$ | $C$ | **Remarks** |
|---|---|---|---|---|
| 8 | 5 | 5 | | Impossible. $A \neq O$ |
| 7 | 6 | 4 | | Impossible. $A \neq M$ |
| 6 | 7 | 3 | 8 | might be possible |
| 5 | 8 | 2 | 7 | might be possible |

If $O + T = 13$, then $L + H > 10$ and $L + H = 17\ or\ 18$. However, there is no solution when $L + H = 17\ or\ 18$ since $L$ and $H$ cannot be 8 or $L = H$. Case 3 is impossible.

Case 4: if $O + T = 14$, then $O + A = 10$. $O + T = 17$ has 2 possible solutions:

| $T$ | $O$ | $A$ | $C$ | |
|---|---|---|---|---|
| 9 | 5 | 5 | | Impossible. $A \neq O$ |
| 8 | 6 | 4 | | Impossible. $A \neq M$ |
| **6** | **8** | **2** | **7** | **possible** |
| 5 | 9 | 1 | | Impossible. $A \neq S$ |

The value of S + A + S + M + O is $1 + 2 + 1 + 4 + 8 = \mathbf{16}$.

Answer: **16**

# Solutions to SASMO 2020 Secondary 4 (Grade 10)

## Question 1

$$2020 \times 999 - 9999 \times 202 + 9999 = 202 \times (9990 - 9999) + 9999 = 9999 - 202 \times 9$$
$$= \mathbf{8181}$$

Answer: **(A)**

## Question 2

$5^{2020} \times 4^{1008} = 5^{2020} \times (2^2)^{1008} = 5^{2020} \times 2^{2016} = 5^4 \times (5 \times 2)^{2016} = 625 \times 10^{2016}$

$625 \times 10^{2016}$ has $3 + 2016 = \mathbf{2019}$ digits.

Answer: **(D)**

## Question 3

Probability(an event) = (# of ways an event can happen) / (Total number of outcomes)

The total number of outcomes when a standard 6-sided dice is rolled twice is $6 \times 6 = 36$.

Find the probability for options A to D:

A. The number of ways option A can happen is 3: (1, 2), (2, 4) and (3, 6). P(A)=$3/36$.
B. The number of ways option B can happen is 21: (1, 1), (2, 1), (2, 2), (3, 1), …, (6, 6). P(B)=$21/36$.
C. The number of ways when both numbers are less than 4 is 9: (1, 1), (1, 2), (1, 3), (2, 1), …, (3, 3). P(C)=$1 - \frac{9}{36} = \frac{27}{36}$.
D. The number of ways option D can happen is 15: (1, 1), (1, 2), (1, 4), (1, 6), …, (6, 5). P(D)=$15/36$.

Option **C** has the highest probability of occurrence.

Answer: **(C)**

## Question 4

There are $6 \times 5 = 30$ players in total. If all 30 players played with each other, then the total number of games would be $30 \times 29 \div 2 = 435$. However, the number of games between players from the same country is $5 \times 4 \div 2 = 10$. Thus, the number of games in the tournament was $435 - 6 \times 10 = \mathbf{375}$.

Answer: **(C)**

## Question 5

A Nicely Seven number must be a prime number. Let $a$ be a Nicely Seven number, then $a$ and $1$ are its factors and $a + 1$ is a multiple of 7.

List down all two-digit multiples of 7, then subtract 1 from them and check whether the results are prime numbers:

| 14 | 21 | 28 | 35 | 42 | 49 | 56 | 63 | 70 | 77 | 84 | 91 | 98 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 13 | 20 | 27 | even | 41 | even | 55 | even | 69 | even | 83 | even | 97 |

Thus, there are **4** Nicely Seven numbers.

Answer: **(D)**

## Question 6

$$\left(2^{\frac{1}{3}}\right)^3 + 1^3 = \left(2^{\frac{1}{3}} + 1\right) \times \left(4^{\frac{1}{3}} - 2^{\frac{1}{3}} + 1\right) \Leftrightarrow 3 = \left(2^{\frac{1}{3}} + 1\right) \times a \Leftrightarrow$$

$$\frac{3}{a} - 1 = 2^{\frac{1}{3}} \Leftrightarrow \frac{3-a}{a} = 2^{\frac{1}{3}} \Leftrightarrow \left(\frac{3-a}{a}\right)^6 = \left(2^{\frac{1}{3}}\right)^6 = \mathbf{4}$$

Answer: **(A)**

## Question 7

*Case 1:* $x > 57$

$$\left|\frac{1}{57-x}\right| < \frac{5}{x^2} \Leftrightarrow \frac{1}{x-57} < \frac{5}{x^2} \Leftrightarrow x^2 - 5x + 285 < 0$$

The inequality $x^2 - 5x + 285 < 0$ does not have any solutions.

*Case 2:* $x < 57$

$$\left|\frac{1}{57-x}\right| < \frac{5}{x^2} \Leftrightarrow \frac{1}{57-x} < \frac{5}{x^2} \Leftrightarrow x^2 + 5x - 285 < 0$$

The solution of the inequality $x^2 + 5x - 285 < 0$ is

$$\frac{1}{2}\left(-5 - \sqrt{1165}\right) < x < \frac{1}{2}\left(\sqrt{1165} - 5\right).$$

The largest possible value of $x$ is **13**.

Answer: **(A)**

## Question 8

Count by the number of 2s in the sum as shown in the table below.

| Number of 2s | Number of 1s | Number of ways |
|---|---|---|
| 4 | 1 | 5 |
| 3 | 3 | 20 |
| 2 | 5 | 21 |
| 1 | 7 | 8 |
| 0 | 9 | 1 |
| **The total number of ways:** | | **55** |

Answer: **(B)**

## Question 9

Since $AE:EC = 1:2$, then $Area(AEF):Area(CEF) = 1:2$ and $Area(AEB):Area(CEB) = 1:2$.

Similarly, $Area(BDA):Area(CDA) = 1:2$ and $Area(BDF):Area(CDF) = 1:2$.

Let $Area(AEF) = a$ and $Area(BDF) = b$, then $Area(CEF) = 2a$ and $Area(CDF) = 2b$.

From $Area(AEB):Area(CEB) = 1:2$, we get $Area(AFB) = 1.5a$

From $Area(BDA):Area(CDA) = 1:2$, we get $Area(AFB) = 1.5b$. Hence $a = b$, $Area(ABC) = 7.5a$ and $Area(AEF):Area(ABC) = a:7.5a = \mathbf{\frac{2}{15}}$.

Answer: **(D)**

## Question 10

Count the number of '6's in hundreds place:

When the hundreds place is 6, there are 10 options (0, 1, 2, …, 9) for tens place and 5 options (0, 2, 4, 6, 8) for ones place. There are $10 \times 5 = 50$ sixes in hundreds place.

Count the number of '6's in tens place:

When the tens place is 6, there are 9 options (1, 2, …, 9) for hundreds place and 5 options (0, 2, 4, 6, 8) for ones place. There are $9 \times 5 = 45$ sixes in tens place.

Count the number of '6's in ones place:

When the ones place is 6, there are 9 options (1, 2, …, 9) for hundreds place and 10 options (0, 1, 2, …, 9) for tens place. There are $9 \times 10 = 90$ sixes in tens place.

There are $50 + 45 + 90 = \mathbf{185}$ **red digits** '6' on the piece of paper.

Answer: **(A)**

## Question 11

From the first equation:

$$\frac{ab}{a+b} = 4 \Leftrightarrow \frac{1}{4} = \frac{(a+b)}{ab} = \frac{1}{b} + \frac{1}{a}$$

Similarly, we get $\frac{1}{6} = \frac{1}{b} + \frac{1}{c}$ and $\frac{1}{12} = \frac{1}{c} + \frac{1}{a}$

Adding the 3 equations, we get

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = \frac{1}{2} \times \left(\frac{1}{4} + \frac{1}{6} + \frac{1}{12}\right) = \frac{1}{4}$$

Then

$$\frac{abc}{ab+bc+ac} = \frac{1}{\frac{ab+bc+ac}{abc}} = \frac{1}{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}} = \frac{1}{\frac{1}{4}} = \mathbf{4}$$

Answer: **(C)**

## Question 12

If $a$ and $b$ are odd numbers, then $a^2$ and $b^2$ give remainder 1 on division by 4 and the sum $a^2 + b^2$ gives remainder 2 on division by 4. However, 2020 gives remainder 0 on division by 4. Hence $a$ and $b$ are even numbers.

Let $a = 2x$ and $b = 2y$ and rewrite the original equation:

$$(2x)^2 + (2y)^2 = 2020 \Leftrightarrow x^2 + y^2 = 505 \quad (1)$$

Obviously, $x$ and $y$ must have different parities.

If $x$ is an even number, then only $x = 8$ and $x = 12$ satisfy the equation (1). Since the equation (1) is symmetric, then there are 4 solution pairs of $(x, y)$ or **4 pairs** of $(a, b)$: (42, 16), (38, 24), (24, 38) and (16, 42).

Answer: **(D)**

## Question 13

Since the segment $BE$ is tangent to the circle, then $AE$ is perpendicular to $BE$ and $AEB$ and $CDB$ are similar triangles.

Let $AE = AC = r$, then by Pythagoras Theorem in triangle $AEB$:

$$AE^2 + BE^2 = AB^2 \Leftrightarrow r^2 + 12^2 = (r+8)^2$$

Solving the equation, we get $r = 5$ and $AB = 13$.

From similar triangles $AEB$ and $CDB$:

$$\frac{AE}{AB} = \frac{CD}{BC} \Leftrightarrow \frac{5}{13} = \frac{CD}{8} \Leftrightarrow CD = \mathbf{\frac{40}{13}}$$

Answer: **(C)**

## Question 14

If Diana and Jason go to Grand Mountain Secondary School, then neither Tom nor Michael goes to Grand Mountain Secondary School. Hence Tom and Michael go to Silver Oak Secondary School which is impossible. Thus, Diana and Jason go to Silver Oak Secondary School.

If Alexa and Michael are Secondary 4 students, then neither Tom nor Jason is a Secondary 4 student. Hence Jason and Tom are Secondary 2 students which is impossible. Thus, Alexa and Michael are Secondary 2 students.

Diana, Jason, Alexa and Michael cannot be Secondary 4 student from Grand Mountain Secondary School. Thus, the answer is **Tom**.

Answer: **(D)**

## Question 15

Let $a_1, a_2, \ldots, a_6$ be the number of marbles given to each of the 6 boys. Then $a_1 \geq 3, a_2 \geq 3, \ldots, a_6 \geq 3$ and rewrite $a_1 = b_1 + 2, a_2 = b_2 + 2, \ldots, a_6 = b_6 + 2$ where $b_1 \geq 1, b_2 \geq 1, \ldots, b_6 \geq 1$.

It is given that

$$(b_1 + 2) + (b_2 + 2) + \cdots (b_6 + 2) = 30 \Leftrightarrow b_1 + b_2 + \cdots b_6 = 18$$

The number of solutions for the above equation is $C(17,5) = \binom{17}{5} = \frac{17!}{5! \times (17-5)!} = \mathbf{6188}$.

Answer: **(A)**

## Question 16

The largest possible sum of the digits of a 4-digit number is $9 \times 4 = 36$. Since the sum of digits is a multiple of 25, then the sum of the digits must be 25. The last 2 digits must be 00, 25, 50 or 75.

Case 1:

If the last 2 digits are 00, then the sum of the digits is at most 18 which is impossible.

Case 2:

If the last 2 digits are 25, then the sum of the hundreds and thousands digits is 18. However, $9 \times 9 \times 2 \times 5$ is not a multiple of 25.

Case 3:

If the last 2 digits are 50, then the sum of the digits is at most 23 which is impossible.

Case 4:

If the last 2 digits are 75, then the sum of the hundreds and thousands digits is 13. For the product to be a multiple of 25, the hundreds or thousands digit must be 5. Thus, the largest possible 4-digit number is **8575**.

Answer: **8575**

## Question 17

$$\frac{3 \times 111.111}{6 \times 1.001} + \frac{3 \times 222.222}{6 \times 2.002} + \cdots + \frac{3 \times 666.666}{6 \times 6.006} =$$

$$\frac{3 \times 111 \times 1.001}{2 \times 3 \times 1.001} + \frac{3 \times 222 \times 1.001}{2 \times 3 \times 2.002} + \cdots + \frac{3 \times 666 \times 1.001}{2 \times 3 \times 6.006} =$$

$$\frac{111}{2} + \frac{111}{2} + \cdots + \frac{111}{2} = \frac{666}{2} = \mathbf{333}$$

Answer: **333**

## Question 18

The six-digit number 21A3B8 is divisible by 3 and 11.

From the Divisibility Rule of 11:

$|(2 + A + B) - (1 + 3 + 8)| = |A + B - 10|$ must be divisible by 11. Hence $A + B = \mathbf{10}$.

Answer: **10**

## Question 19

$$\angle ADE = 90° - \angle EAD = 90° - \angle CAB = \angle BCA = \angle GCF$$

$$\angle EAD = 90° - \angle ADE = 90° - \angle GCF = \angle FGC$$

Thus, triangles $AED$ and $GFC$ are similar and

$$\frac{AE}{DE} = \frac{GF}{FC} \Leftrightarrow \frac{8}{12} = \frac{12}{FC} \Leftrightarrow FC = 18.$$

Quadrilateral $ACGD$ is trapezium and its area is

$$DE \times \frac{(GD + AC)}{2} = 12 \times \frac{(12 + 38)}{2} = \mathbf{300}.$$

Answer: **300**

## Question 20

Clearly, the desired 2-digit number must be an even number. The numbers 98, 96, 94, …, 72, 70 can be written as the sum of 2 different prime numbers in more than 2 different ways. For example,

| | | | |
|---|---|---|---|
| **98** | $79 + 19$ | $31 + 67$ | $37 + 61$ |
| **96** | $7 + 89$ | $13 + 83$ | $17 + 79$ |
| **94** | $5 + 89$ | $11 + 83$ | $23 + 71$ |
| … | | | |

The greatest 2-digit number that can be written as the sum of 2 different prime numbers in exactly 2 different ways is $\mathbf{68} = 31 + 37 = 7 + 61$.

Answer: **68**

## Question 21

The number must have $6 + 12 = 18$ divisors. Then its prime factorisation must be $p^2 \times q^2 \times r,\ p \times q^8, p^2 \times q^5$or $p^{17}$. Since the number has exactly 6 odd divisors, then its prime factorisation must be $p^2 \times q^2 \times r$ or $p^2 \times q^5$. The smallest such number is $2^2 \times 3^2 \times 5 = \mathbf{180}$.

Answer: **180**

## Question 22

$$x^2 + y^2 + x^2y^2 + 10 - 6x - 6y = 0 \Leftrightarrow$$

$$x^2 + y^2 + x^2y^2 + 10 - 6x - 6y + 2xy - 2xy = 0 \Leftrightarrow$$

$$x^2y^2 - 2xy + 1 + x^2 + 2xy + y^2 - 6(x + y) + 9 = 0 \Leftrightarrow$$

$$(xy - 1)^2 + (x + y)^2 - 6(x + y) + 9 = 0 \Leftrightarrow$$

$$(xy - 1)^2 + [(x + y) - 3]^2 = 0$$

Hence $xy = 1,\ x + y = 3$ and

$$3^2 - 2 \times 1 = (x + y)^2 - 2xy = x^2 + y^2 = \mathbf{7}.$$

Answer: **7**

## Question 23

To get the least number of weights, we need to put the weights on both sides of a scale.

To weigh 1 g object, we need 1 g weights.

To weigh 2 g object, we need 2 or 3 g weights.

To weigh 5 g and 14 g objects, we need at least 2 more weights. Thus, the least number of weights is **4** and the weights are 1 g, 3 g, 9 g and 27 g. For example,

5 g = 9 g – 3 g – 1 g, 14 g = 27 g – 9 g – 3 g – 1 g and 35 g = 27 g + 9 g – 1 g.

Answer: **4**

## Question 24

Rewrite the sum as the product:

$$\begin{array}{cccccccc} & C & I & R & C & L & E \\ \times & & & & & & 3 \\ \hline & S & P & H & E & R & E \end{array}$$

In ones place, the letter E must be 5. (E=0 can be checked similarly, and it doesn't lead to any solution.)

The six-digit number CIRCLE times 3 results in a 6-digit number, therefore C=1, 2 or 3.

If C=3, then in hundreds place product $C \times 3$, there must be a carry over of 6 from tens place product which is impossible.

If C=2, then in hundreds place product $C \times 3$, there must be a carry over of 9 from tens place product which is impossible. Hence C=1.

In hundreds place product $C \times 3$, there must be a carry over of 2 from tens place product which implies that L=8 or 9.

If L=8, then R=5=E which is impossible. Hence L=9, R=8 and H=4.

In hundred thousands place product, S cannot be 4 and 5 since H=4 and E=5. Hence S=3 and the only one-digit number left for I is 0.

The value of the sum C + I + R + C + L + E is $1 + 0 + 8 + 1 + 9 + 5 = \mathbf{24}$.

Answer: **24**

## Question 25

The total number of ways to place all his books without any restrictions is $7! = 5040$.

The number of ways when the Physics book is next to the Biology book and Geometry is next to the Chemistry one is $5! \times 2 \times 2 = 480$.

Hence, the number of ways when the Physics book is not next to the Biology book and Geometry is not next to the Chemistry one is $5040 - 480 = \mathbf{4560}$.

Answer: **4560**